

MUHAMAD ILHAM FADILLAH SUHERLAN

SYSTEM ANALYST | SOFTWARE ENGINEERING FRESH GRADUATE

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SUMMARY

Software Engineering fresh graduate with hands-on internship and academic project experience in requirements gathering and elicitation, existing-system analysis, system modeling, database design, technical documentation, and software testing. Experienced in translating stakeholder needs into functional and non-functional requirements and supporting implementation through system/data models, UAT, integration testing, validation testing, SQL, PostgreSQL, and REST API.

EDUCATION

Diploma (D3) in Software Engineering | Politeknik Negeri Bandung - Kabupaten Bandung Barat, Indonesia | Aug 2023 - Aug 2026 | GPA 3.58/4.00

TECHNICAL SKILLS

Requirements & System Analysis: Requirements Gathering, Requirements Elicitation, Stakeholder Identification, Stakeholder Interview, Existing-System Analysis, User Requirements, Business Rules, Functional Requirements, Non-Functional Requirements, Use Case Analysis, Requirements Validation, Technical Documentation

System Modeling & Design: System Architecture, UML, Use Case Diagram, Sequence Diagram, Class Diagram, Package Diagram, Domain Model, Deployment Diagram, ERD, Data Design, DFD, UI Design

Database & Testing: SQL, PostgreSQL, Database Design, REST API, UAT, Test Case Development

Tools & Technical Exposure: PHP, Laravel, Dart, Kotlin, Python, C, C++, Java, Spring Boot, Git, GitHub, Figma, Postman, PowerDesigner, Draw.io, PlantUML

WORK EXPERIENCE

Balai Bahasa Provinsi Jawa Barat | Bandung, Jawa Barat

Intern – System Analyst & Mobile Developer | Jul 2025 – Oct 2025

- Conducted stakeholder interviews and existing-system analysis; translated 2 use cases and 3 sub-use cases into 9 functional and 4 non-functional requirements.
- Designed the ERD and UI.
- Implemented the main flipbook feature (Dart) of Kado Ceria, a bilingual (Sundanese–Indonesian) interactive audio storybook app for early childhood, BIPA learners, and persons with disabilities.

PROJECT EXPERIENCE

CONTENT MANAGEMENT SYSTEM - KADO CERIA | <https://github.com/DAlIRIJTK/cms-kadoceria>

Project Lead | System Analyst & Technical Writer | Backend Developer | Quality Assurance | Laravel, PHP, PostgreSQL, REST API, AWS
Jan 2026 – Aug 2026

- Analyzed the existing hardcoded Kado Ceria mobile app, where every flipbook content update required code changes and redeployment, and interviewed Balai Bahasa Provinsi Jawa Barat to define content management requirements.
- Elicited 10 user requirements and 14 business rules; elaborated them into 34 functional requirements and 10 non-functional requirements across 2 use cases.
- Designed system architecture, package diagram, 12 sequence diagrams, class diagram, deployment diagram, and conceptual/logical/physical data models.
- Prepared 44 UAT test cases and validated requirements through integration, black-box, and post-deployment NFR testing (43 test cases run before and after deployment).

MOBILE APPLICATION – GROW | <https://github.com/grow-app-4/grow-app-mobile>

System Analyst | Mobile Developer | Kotlin, PostgreSQL
Feb 2025 – Jun 2025

- Identified stakeholders and designed the domain model; translated 5 use cases and 1 sub-use case into 9 functional and 7 non-functional requirements.
- Designed the ERD/database, implemented daily intake monitoring, and developed 10 test cases with a 100% pass rate.

ADDITIONAL TECHNICAL PROJECTS

Suku Face Recognition | <https://github.com/yanirahmawati234/suku-face-recognition>

Backend Developer | Python
Feb 2025 – Jun 2025

- Refactored preprocessing using MTCNN

Gamification - Semaphore | https://github.com/ilhamfs7/TUBES_PCD_PR_Gamifikasi_Semaphore

Backend Developer | Python
Feb 2025 – Jun 2025

- Modified detection logic and input vs target comparison

Hidingdata | <https://github.com/ilhamfs7/proyek2-kel2>

Backend Developer | C
Feb 2024 – Jun 2024

- Implemented LSB encryption/decryption with a doubly circular linked list

Search Application Using Trie Data Structure | <https://github.com/Yorubreak/Tugas-Besar-SDA>

Backend Developer | C++
Feb 2024 – Jun 2024

- Implemented Trie-based search

CERTIFICATION & TRAINING

Arutala Junior Backend Developer using Spring Boot – Distinguished | Arutalalab | July 2026

Test of English for International Communication (TOEIC) | Politeknik Negeri Bandung | Score: 765 | May 2026

Basic Management Skills Training (LKMM-TD) | Politeknik Negeri Bandung | Nov 2023